

Physics

Q1

→ The light beam will be real and inverted.

Q2

→ $f = \frac{R}{2}$ [Radius of curvature = 50]

$$f = \frac{50}{2} = 25$$

$$f = 25$$

Focal length = 25

It will be a convex mirror

Q3

It will be a convex mirror

$$\frac{v}{u} = m$$

$$2.5 = m$$

It will be a convex mirror

Physics

AQ4

$$h_2 \rightarrow 2 \text{ cm}$$

$$f \Rightarrow 10 \text{ cm}$$

$$u \Rightarrow 15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{10} = \frac{1}{v} + \frac{1}{15}$$

$$\frac{1}{10} - \frac{1}{15} = \frac{1}{v}$$

$$\frac{3}{30} - \frac{2}{30} = \frac{1}{v}$$

$$\frac{1}{30} = \frac{1}{v}$$

$v = +30$ $\rightarrow$ 30 cm from the mirror is the position of image

$$m = \frac{+v}{u}$$

$$m = \frac{30}{15}$$

~~mag~~ magnification ~~size~~ $\Rightarrow 2$
size is ~~smaller~~ magnified because $m = 2$
~~and~~ $m > 1$

~~Set~~ Nature $\Rightarrow$ Virtual and erect.

Q1

A $\Rightarrow$

Reaction to 40°

Exom

AQ2 $\rightarrow$ a]

b]

AQ3 a]

b]

Zinc SO